

5(a)	Hooke's (law)	B1
5(b)(i)	$\sigma = F / A$	C1
	$= 36 / (4.1 \times 10^{-7})$	A1
	$= 8.8 \times 10^7 \text{ Pa}$	
5(b)(ii)	Young modulus = σ / ϵ or $F / A\epsilon$	C1
	$\epsilon = 8.8 \times 10^7 / (1.7 \times 10^{11})$	A1
	$= 5.2 \times 10^{-4}$	
5(c)	$R = \rho L / A$	C1
	$\Delta R = \rho \Delta x / A$	C1
	$= 3.7 \times 10^{-7} \times 0.12 \times 10^{-3} / (4.1 \times 10^{-7})$	
	$= 1.1 \times 10^{-4} \Omega$	A1
5(d)	remove the force/ F and wire returns to original length	B1